

Do Targeted Discount Offers Serve as Advertising? Evidence from 70 Field Experiments

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Abstract. The prevalence and widespread usage of email has given businesses a direct and cost-effective way of providing consumers with targeted discount offers. While these discounts are expected to increase the demand for the promoted products, are they effective in increasing revenues? Do they have effects beyond acting as price reductions? We study these questions using individual-level data from 70 randomized experiments run by a large online ticket resale platform. We estimate the redemption rates of the offers and also measure the broader impact of emailed promotions by comparing purchases by individuals who received the experimental promotions with purchases by those who did not receive the offers because of the experimental randomization. We find that the offers cause the average expenditure to increase significantly, by \$3.03 (a 37.2% increase) during the promotion window. However, the redemption rate of these offers is low. Importantly, 90% of these gains are not through redemption of the offers. The individuals who spent more on the platform in the past are more responsive to the offers, and the effect of the offers is significantly higher among individuals who did not transact on the platform in the year before the offer was given. Interestingly, the offer causes carryover to the week after the promotion expires; we find that spending increases by \$1.55 in the week after the offer expires. Additionally, we find evidence for cross category spillovers to nonpromoted products: offers not applicable to a ticket genre cause an increase in spending in that genre. We conclude that emailed offers can serve as a form of “advertising” for the firm’s products, in addition to being tools for price discrimination.

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1. Introduction

Being able to target individual consumers is one of the most important objectives of marketing in general and of digital marketing in particular. In the offline world, firms have been using targeted coupons based on consumers’ demographic characteristics or their actual purchase behavior. In the online world, email marketing has emerged as a channel that allows targeted communication with customers. In a recent survey, 99% of marketers reported investing in it (Experian Marketing Services 2014). Businesses widely use emails to target consumers with customized coupons or discounts (Banjo 2014). While there has been research on issues such as how the nature of content in an email influences consumers’ responses (Ansari and Mela 2003, Wattal et al. 2011), questions related to targeting and the direct impact of email discounts/offers on redemptions, revenues, and profits have not been much researched. This paper aims to study these issues.

How should we think about targeted emailed offers? One could view them as vehicles for promotional discounts or coupons. Traditionally, such targeted

discounts have been thought of as tools for price discrimination. Therefore, according to this view, a discount offer appeals to the price sensitivity of the individual receiving it, thereby increasing the recipient’s chances of buying the promoted product. Hence, the firm giving the offer would want to target price-sensitive individuals who would have not bought the product otherwise. In this case, the success of an email offer campaign would depend on how many people actually use the offered discounts. However, the effect of the offers may go beyond the role of the coupon embedded in the email. In particular, an email with a discount may play a role similar to that of an advertising message.¹ The offer in an email can draw the recipient’s attention and register the firm’s products in the consumer’s memory, which can make her more likely to consider the firm’s products on the next purchase occasion. This view has distinct implications for firms. First, they might not want to target the offers specifically to price-sensitive individuals. Such offers would be targeted to individuals who might be unaware or might have forgotten the products the firm sells. Second, the number of individuals who actually use the offer does

not capture the complete impact of the email campaign. The effectiveness of the offer email has to be judged on other metrics, such as total sales. The true impact of an offer could lie somewhere on the spectrum between a “pure price reduction effect” and a “pure advertising effect.”

Motivated by these contrasting possible outcomes of targeted email offers, we focus on the following empirical questions. The first question relates to quantifying the overall effectiveness of targeted online offers. Can they be profitable for a firm? Second, we ask whether the effect of the offers can go beyond their effect as price reductions. Can the offers serve as advertising messages and increase spending even for individuals who do not redeem the offers? Is the advertising effect of the discount offers statistically and economically significant? Do the offers cause carryover and spillover effects as noted in the past research on advertising?

Measuring email campaign effectiveness is complicated by the very nature of email marketing; i.e., it is targeted. Estimating incremental sales via a simple comparison of outcomes across those receiving and not receiving the email is rendered inappropriate by the selection inherent in targeting. Well-designed experiments can, in theory, overcome this problem. In practice, however, these experiments need to be executed with care because they can be nonrepresentative, costly, and time consuming. A second challenge emerges in estimating the effects of emails. Because emails are easily customizable, each campaign may comprise a small set of targeted individuals. Therefore, each campaign, by itself, may have insufficient observations to be able to precisely measure its impact on a firm's revenues.

We address these issues by reporting average effects across 70 experiments conducted by a large online ticket resale platform. Each of these experiments is a targeted email offer campaign run by the platform during the time period from July 2009 through December 2011. These campaigns are representative of the targeted discount promotions the platform offered to encourage repeat buying on the website among customers who did not opt out of email communication. The data available to us are from a panel of these customers who made purchases through the platform. While the email offers are themselves targeted based on various criteria used by the platform, for each of these targeted offers, a random subset was “withheld” from actually receiving the offer. Importantly, we observe the targeted group for each offer as well as the treated subset (the group that actually received the email) and the control subset (the group for which the email promotion was withheld). For this reason we are able to causally link the emailed offers with observed differences in purchases between treated and control conditions.

In our analysis, we find that email offers have a positive causal effect on expenditures on the platform.

Further, the effect is statistically and economically significant. On average, receiving experimental offers increases a consumer's spending by approximately \$3 (a 37.2% increase on a base of \$8.07). Importantly, a very small portion of the gains come from the actual redemption of the offers. Approximately 90% of the gains from experimental offers is through increased spending without using the discounts offered. If the discount aspect of the offers were the main driving factor, we would have seen very little increase beyond what is justified by the redemption of the offers, which is not the case in our data. Therefore, our finding suggests that the experimental offers have effects beyond merely acting as price reductions.

We find that the impact of the offers is highest among individuals who spent more on the website before the offer was given out. Interestingly, the offers are significantly more effective in increasing spending by individuals who had not transacted on the website recently, compared to individuals who had transacted on the website within a year of getting an offer. This finding is consistent with the reminder effect of the offers: viewing an offer from the firm increases the chances of the individual getting reminded of and considering the firm's offerings for purchase.

We further investigate whether the targeted email offers show carryover (over time) and spillover (to non-promoted categories) effects as if the emails acted as advertising, as described previously. We find that the offer emails have a significant effect during the week after the offer expires, increasing expenditure by \$1.55 on average, although not beyond. Additionally, we find that the promotions cause increased sales of products for which the discounts do not apply. Specifically, we use the fact that many offers in our data are not applicable on purchases of Major League Baseball (MLB) tickets. For these offers we find that that expenditure on MLB tickets increases significantly during promotions although the discounts are not applicable.

These findings enable us to make several contributions to the literature. This paper shows evidence for significant causal effects of email offers on sales. Given the nature of this instrument, the prior belief marketers might have about such offers is that they serve as a mechanism for price discrimination. However, this paper shows that a substantial portion of the gain from the offers arises even when the offers are not redeemed by the recipients. This suggests that email offers may serve as “advertising” for the product and for the firm (in our case, the resale platform), in addition to serving as tools for price discrimination. Consequently, the firm might consider targeting individuals who may have forgotten about its products, rather than just sending the offers specifically to price-sensitive individuals. If the advertising effect dominates, as in our case, the relevant trade-off firms face is the gain in sales due to

emails versus the cost of designing and sending the emails, the (possible) opportunity cost of customers paying less attention to future emails, etc. The firm benefits from avenues not confined by the characteristics of the offer; sales increase even after the offer expires and through purchases where the offer is not even applicable. In other words, advertising-like effects can exist even for marketing actions that are not explicitly ads. This finding suggests that focusing only on redemption rates of email promotions is not enough to gauge the impact of email offers. These results have direct implications for firms investing in email marketing and also policy makers interested in evaluating the impact of prevalent targeted offers. They also raise questions (e.g., is a discount even needed as part of the email, does its presence encourage consumers to open the email, etc.) that future research can help to answer.

The rest of this paper proceeds as follows. In the next section we provide a brief review of the relevant literature. This is followed by a section on the specific empirical setting and the data we use. Next, we provide a brief description of our empirical measurement approach. Following that we discuss our findings and then conclude with a brief discussion.

2. Literature Review

Past research on email marketing is limited and does not directly address our research questions. Lewis (2004) estimates a dynamic demand model to study the effects of a loyalty program and compares it with email promotions. Hartmann (2006) studies consumption utility in a context where email promotions are used to manage excess capacity. Kumar et al. (2014) study the impact of marketing activities on the propensity of a consumer getting in and out of email marketing lists. Our paper adds to the literature by providing evidence of causal effects of email offers on sales using field experiments and describing how these effects can impact a firms' targeting strategy.

More broadly, this paper relates to the extensive research literature that investigates how discounts and coupons affect consumer decisions.² Blattberg et al. (1995) summarize the early research in this area. Coupons have been primarily thought of as a tool for price discrimination (Narasimhan 1984, Banks and Moorthy 1999, Lu and Moorthy 2007), allowing firms to charge a lower price to consumers who are willing to spend effort on redeeming the coupons and are likely to have a lower willingness to pay. Many studies on the characteristics of individuals who redeem coupons support this view (Bawa et al. 1997, Cronovich et al. 1997, Swaminathan and Bawa 2005, Chiou-Wei and Inman 2008).

A large section of the empirical work in this area has focused on the impact of coupons on the promoted brand's sales. Klein (1981) reports a field experiment

where a random subset of the target population does not receive a coupon and finds that coupons increase sales for the promoted brand. Other research efforts estimate models of consumer demand incorporating the effects of coupons. Neslin (1990) uses household panel data to estimate a market share model and finds that the increase in sales associated with coupons is not large enough for couponing to be profitable for the firms considered. Chiang (1995) estimates a demand model allowing for category expansion due to coupons and finds no evidence for such an effect in detergent purchases. Gönül and Srinivasan (1996) estimate a structural model to find evidence that consumers have expectations about receiving coupons and incorporate them into their decision making. Another related section of the literature focuses on the impact of the characteristics of coupons on their effectiveness (Raju et al. 1994). Srinivasan et al. (1995) use data on store sales to find that newspaper freestanding inserts (FSIs) with print ads increase store sales. Leclerc and Little (1997) use lab studies and panel data analysis to find that the execution of print ads along with FSIs matters; response from less brand-loyal consumers increases with the informational content of the ads.³

This paper adds to the above literature by (1) using data from randomized field experiments to credibly measure the causal effects of email offers and (2) shedding light on several aspects of the mechanism of their effect (such as cross category spillovers), which have not been considered in the previous literature. Since marketing communications, such as the offers considered in this paper, are highly targeted, a randomized control group is necessary for unbiased estimation of their causal effects on consumer purchases. In the absence of a randomized control group, comparison of purchases before and after the distribution of offers might yield biased estimates because offers could co-occur with other promotions or other incidental factors correlated with buying behavior.⁴ For example, Nevo and Wolfram (2002) and Anderson and Song (2004) both show that coupons and price reductions are coordinated. Further, an experiment is especially useful in credibly identifying the mechanism driving the effects. For example, to credibly show the advertising effects of discount offers, it is important to show that coupons increase sales for individuals who do not redeem them. In the absence of an experiment, one would compare (a) individuals who are given coupons but do not redeem coupons with (b) those who do not receive coupons. However, groups (a) and (b) are likely to be different in baseline preferences and purchase trends because of selection by the marketer while targeting the coupons and because group (a) self-selects itself by not redeeming the coupons. Therefore, this empirical strategy would be problematic. However, because of experimentation, our data set is unique and allows us

to overcome such problems by comparing randomized treatment and control groups.

Closest in this spirit to our paper is the study by Klein (1981). However, that paper reports a field experiment that estimates the effect of coupons on sales of the promoted brand and considers neither the advertising mechanism as we do nor the long-term effects of coupons. In a similar vein, Bawa and Shoemaker (1989, 1987) find that, compared to before sending the coupons, purchase for the promoted brand increases even among individuals who did not redeem the coupons. Venkatesan and Farris (2012) use panel data to estimate a model of consumer demand to show how customized targeted coupons affect sales in a supermarket. Similar to Bawa and Shoemaker (1989), they find that consumers who receive but do not redeem the coupons spend more than those who do not receive them. Our paper adds to this research in several ways. First, because our data are generated from randomized experiments, we are able to alleviate any selection concerns mentioned in the previous paragraph. This step is important for credibly establishing the mechanism through which email offers work. Second, we are able to leverage the panel nature of our data to more deeply understand the mechanism driving the effects. We are able to correlate the causal effects of the offers with customers' characteristics such as the recency, frequency, and monetary value of past purchases and suggest a suitable targeting strategy. Third, we discuss and tackle the econometric challenges in combining several experiments, which may get overlooked while estimating a model, as previous research has had to do in order to deal with the selection problems.

Our paper finds that consumers leave substantial gains on the table, perhaps attributable to forgetting or to suboptimal decision making (Gabaix and Laibson 2006, Brown et al. 2010, De Clippel et al. 2014). In our context, these gains left out by the consumers benefit the firm significantly.

To summarize, our paper contributes to the targeted coupons/promotions literature by uncovering causal evidence on the effects of such promotions in the presence of targeting. Additionally, it sheds light on the advertising role of these promotions, which is beyond the traditional role of providing a price discount. Given this advertising role of the email campaigns, we show the presence of spillovers over time (longer-term effects) and across other nonpromoted products.

3. Empirical Setting and Data

3.1. Empirical Setting

The data for this study come from a large online ticket resale platform for the United States and Canada, where millions of tickets get sold annually.⁵ To preserve anonymity, we cannot disclose the name of the

firm. Typical sellers in this marketplace are individuals reselling one or more tickets for events such as concerts or sporting events. Sellers set their own prices for and reveal attributes (e.g., venue location, date, etc.) of each ticket they post. Subsequently, potential buyers search and review these prices and ticket attributes and decide whether or not to purchase a ticket. If a transaction occurs, the buyer pays the platform the price of the ticket, the platform takes a cut (on the order of 10% of the listed price from the buyer and approximately the same from the seller), and the seller receives the remainder. All else being equal, the platform profits from more and/or larger transactions. Note that we only observe transaction outcomes, i.e., whether or not a purchase occurred and for how much, but not the search process in which consumers engage. Further, there is no "social" aspect to the email offers; i.e., recipients cannot forward the emails to friends for them to redeem the promotions.

To retain buyers, the platform uses email promotional offers (hereafter "offers") extensively.⁶ A first-time buyer, upon creating an account, specifies his preferences over event genres and elects to opt into or opt out of receiving emails about events and offers. We focus on individuals who opt into receiving emails; they comprise approximately 50% of a random sample of individuals who have completed a transaction on the website in the past. If a user visits the platform's website and receives an offer, the website may display to her a banner highlighting the offer.

Table 1 describes a few of the observed offers. All offers are valid for a fixed time period when the offered discount is available. To use a discount offered, the user has to actively input a code corresponding to the offer while completing the transaction on the website. The offers are targeted; each offer is sent to a subset of the

Table 1. Examples of Promotional Offers

Starting date	Ending date	Description of offer
7/1/2010	10/1/2010	You have three offer codes in your account, each good for free delivery off of any transaction.
8/13/2010	9/13/2010	Spend \$200 or more and get free delivery (excluding MLB tickets). OR spend \$300 or more and get 10% off (excluding MLB tickets). OR spend \$400 or more and get 15% off (excluding MLB tickets).
9/9/2010	10/10/2010	Purchase one time in the next 30 days and get 10% off (excluding MLB tickets). Purchase a second time within 30 days and get 15% off (excluding MLB tickets).
1/28/2011	2/11/2011	5% off offer group.

Note. Offer descriptions provided by retailer.

population typically segmented based on characteristics of the recipients. The nature and the extent of discount varies across the offers: some offer free shipping of tickets, and others provide percentage discounts. The usage of some offers is restricted: they may be applicable to tickets of a certain genre or they may also be associated with a spending threshold (e.g., on purchases of \$100 or more). Because each offer differs in its type of discount and its restrictions on usage, the offers in our data lie somewhere in the range between traditional offline coupons that are specific to one “product” (e.g., \$0.50 off a box of Cheerios cereal) and general retailer discounts that are not (e.g., CVS offering a 20% discount on purchases of \$20 or more).

3.1.1. Experiment Design. This setting is especially unique because the platform ran a large subset of its promotional offers as experiments. Just as any offer, the experimental offers were targeted, possibly based on individuals’ characteristics and past buying behavior. The number of individuals targeted with the offers also varied. However, within the pool of individuals targeted for an experimental offer, individuals were randomly partitioned into a group that actually received the offer (treatment) and a group that was excluded from receiving the offer (control). We define a campaign to be any given treatment/control pair of conditions. We refer to a campaign as an experiment or an experimental campaign when it has both a treatment and a control group.

3.2. Data

Our data comprise a sample of 95,753 individuals from the population of the platform’s buyers.⁷ Of these, we analyze the 52,043 buyers who opted into receiving emails. This sample of individuals was randomly drawn from a subpopulation of the buyers with accounts on the platform whose first purchases occurred between January 2007 and December 2010.

We observe the campaigns every individual in our sample was targeted with, as well as whether or not the individual received an offer or was excluded (as part of the control group for that email offer).⁸ Hereafter, we refer to individuals who were in an experiment as “targeted” individuals, regardless of whether they were treated or untreated. The features associated with the transactions and transacted items for all of the buyers in our sample are also observed. In particular, we observe the time and date the buyer transacted, the genre of the event purchased, the quantity of tickets purchased, the buyer’s expenditure, and whether the buyer redeemed an offer for this transaction. For a large subset of the individuals (82% of the total), we also observe the demographics of the population residing in their areas (through matched census data; we do not observe zipcodes or any information about where the individuals live). As a rough idea of the size of these

Table 2. Statistics About Expenditure by Individuals in the Data, and the Demographics of the Areas Where They Are Located

	All users	Opt-in users
Number of transactions/year	0.461 (1.546)	0.490 (1.684)
Expenditure/year (\$)	125.09 (955.30)	135.12 (941.56)
Number of tickets/year	1.255 (4.616)	1.332 (5.028)
Region population	19,896.5 (16,420.3)	19,945.3 (16,536.7)
% Black	3.50 (3.95)	3.51 (3.96)
% Nonfamily	9.93 (18.24)	9.03 (17.75)
Avg. family size	2.915 (0.319)	2.930 (0.319)
Avg. HH income (\$)	48,311.9 (32,908.5)	46,850.3 (32,746.3)
Total number of offers	1.049 (1.098)	1.297 (1.154)
Total number of experiments	1.136 (1.097)	1.383 (1.136)
Total number of experimental offers	0.980 (1.024)	1.190 (1.062)
N	95,753	52,043

Notes. The table shows means and standard deviations in parentheses and Compares all buyers with buyers who opted in for receiving emails. HH, household.

transactions, the average transaction size in the data is \$284 with a standard deviation of \$458.⁹ Table 2 reports summary statistics comparing all buyers with those who did not opt out of receiving emails. It appears as though consumers who receive emails are similar to the general population. However, the individuals who opted out are different from those who do not in terms of key observable characteristics such as expenditure and transactions per year (p -value < 0.01). This comparison highlights the need for randomization to create statistically similar groups to accurately estimate the causal effects of the emailed offers.

We observe information on a total of 122 campaigns. Figure 1 shows the distribution of the length of the time window in which the offers are valid; most of the offers expire either in two weeks or in a month. Of the campaigns, 76 are experimental, i.e., have a treatment

Figure 1. (Color online) Distribution of the Number of Days the Offers Are Valid

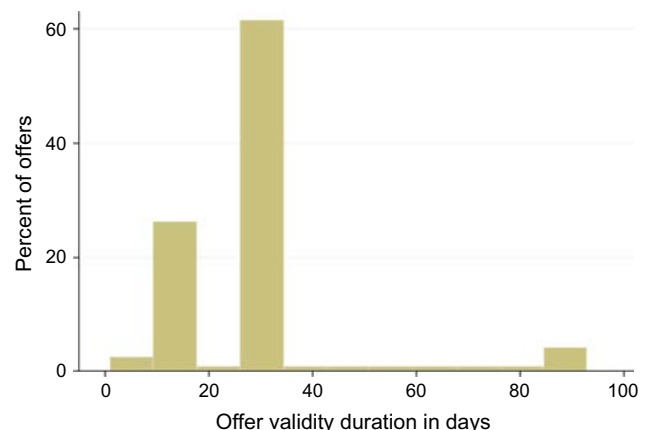


Figure 2. (Color online) Distribution of the Number of Times Targeted (Assignment to Treatment or Control Groups) Across Opt-in Buyers in the Data

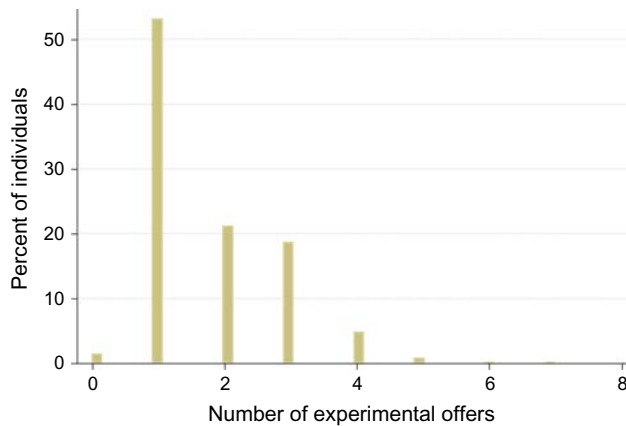


Table 3. Distribution of the Number of Observations Across Experiments

Number of individuals in the experiment	Number of experiments	Treatment probability (%)		
		Mean	Min	Max
<100	6	68	0	91
100 to 200	20	85	33	93
200 to 300	11	82	31	91
300 to 1,000	16	85	47	91
1,000 to 1,500	13	87	49	93
>1,500	10	86	49	97

and a control group. Figure 2 shows the distribution of individuals by the number of experimental campaigns they are exposed to over the course of the data. More than 50% of the individuals are a part of only one experiment, with a majority of the remaining individuals being a part of between two and four experiments. Table 3 shows the distribution of the number of targeted individuals across experimental campaigns. Six of the experiments contain fewer than 100 individuals, approximately half of the experiments contain between 100 and 300 such individuals, and the distribution is skewed with a few experiments with more than 1,500 individuals.

3.3. The Platform's Targeting Strategy

For this study we focus on the 70 experimental campaigns that targeted more than 100 individuals. Using data on a random sample of the buyers, we describe the targeting strategy of the company to answer the following question: who among the users in this platform's database is more likely to get an offer?

Firms are known to target marketing communication based on aspects of an individual's transaction

history. Therefore, for every possible combination of an individual and an offer (whether or not the individual was targeted with the offer), we regress an indicator of whether an individual receives the offer on the individual-specific measures of recency, frequency, and monetary value (RFM measures) at the time the offer was sent, specifically, (r) the number of days since the individual last transacted, (f) the number of transactions made before the offer, and (m) the monetary value of the transactions made prior to the offer. We create dummy variables indicating ranges of these factors to allow for nonmonotonic relationships. We also include demographic characteristics of the three-digit zip code where the individuals live as the explanatory variables in the regression. Table 4 shows estimates of this regression. Note that individuals who have made more transactions in the past are more likely to receive an offer. Similarly, individuals who made larger transactions in terms of dollar value are more likely

Table 4. Regressing an Indicator of an Individual Being Targeted with an Offer on Observed Individual Characteristics

	DV: (0/1) Targeted with an offer			
	I		II	
2 ≤ Frequency of past transactions ≤ 3	0.002**	(0.0002)	0.010**	(0.0005)
4 ≤ Frequency of past transactions ≤ 9	0.006**	(0.0003)	0.019**	(0.001)
10 ≤ Frequency of past transactions	0.120**	(0.0008)	0.028**	(0.002)
Total value past transactions: 2nd quartile	0.0015**	(0.0001)	0.0003	(0.0006)
Total value past transactions: 3rd quartile	0.0029**	(0.0001)	0.0028**	(0.001)
Total value past transactions: 4th quartile	0.0058**	(0.0002)	0.0049**	(0.001)
Days since last transaction: 2nd quartile	0.0098**	(0.0002)	0.0095**	(0.0002)
Days since last transaction: 3rd quartile	0.0047**	(0.0001)	0.008**	(0.0003)
Days since last transaction: 4th quartile	0.0028**	(0.0001)	0.0082**	(0.0004)
Standardized				
Population	−0.0003**	(0.00006)		
% Black population	0.0002**	(0.00006)		
Avg. family size	0.0006**	(0.00006)		
Avg. HH income	−0.00066**	(0.00006)		
Intercept	0.0038**	(0.00011)		
Fixed effect for every individual	No		Yes	
N	95,753 × 70		95,753 × 70	

Notes. The characteristics include (1) frequency of transactions made before the offer, (2) monetary value of the transactions made before the offer, (3) recency of the last transaction, and (4) standardized demographic characteristics of the three-digit zip code where the individuals live. Standard errors in parentheses are robust and clustered by individual. DV, dependent variable; HH, household.

** $p < 0.05$.

to receive offers. As the time since the last transaction increases, the probability of receiving an offer first increases (second quartile has the largest coefficient) and then decreases. Therefore, an individual is most likely to receive an offer after some time (specifically, six months) of making a purchase. Then, as the time since the last transaction increases, the probability of receiving an offer decreases. The association between the demographics of the individual's location and getting targeted with an offer is weaker than the RFM measures. Controlling for the RFM measures, the offers tend to go to individuals who live in lower income areas with larger family sizes and a larger presence of minorities than average.¹⁰

Column II of Table 4 shows that the relationship between receiving an offer and the RFM variables remains similar even if the differences across individuals are controlled for, using fixed effects for each individual. Note that these effects are separately identified since the same household can be targeted with multiple offers.

3.4. Randomization Checks

In total we have 71,445 instances of opt-in individuals being targeted by an experimental campaign. Because of the experimentation, 61,377 instances actually received an offer, and the rest corresponded to instances in the control condition. If randomization were correctly implemented, the probability of getting allocated to the treatment group within an experimental campaign would not be correlated with individual characteristics and past transactions. Table 5 shows the average characteristics including the number of past transactions, total past expenditure, number of previous offers redeemed, and the number of past campaigns targeted for individuals in the treatment and control groups. It shows that the groups are similar based on the observable data. We are unable to

reject the null hypothesis that the average characteristics are the same for both the groups (p -value = 0.33, using Bonferroni adjustment). In another check, we regress a dummy indicator of whether an individual was allocated to the treatment condition on the observed characteristics. Since the check is for systematic variation in within-campaign treatment probability, we add 70 campaign fixed effects to the regression. Estimates show that none of the coefficients related to demographic or past transactions are individually or jointly statistically significant (p -value = 0.6). This analysis supports the firm's contention (and thus our assumption) that, within an experimental campaign, an individual's allocation to the treatment versus control condition is random.

3.5. Redemption of the Offers

Both marketing research and practice considers the rate of redemption (number of offers redeemed/number of offers sent) as the prime measure of success for targeted offers and coupons (Bawa 1996, Musalem et al. 2008, Danaher et al. 2015). Table 6 shows statistics on experimental offer redemption. In total, we observe 61,377 instances where the experimental offers were sent to an individual in the treatment group. In this group 1,475 transactions occur and just 57 involve redemption of the offers. This implies a low redemption rate of $57/61,377 = 0.1\%$ (confidence interval: $[0.077\%, 0.124\%]$).¹¹ These transactions where offers are redeemed account for a total expenditure of \$20,690.

For each instance of an offer being redeemed, we use the redeemed offers' descriptions to compute the discount given.¹² In total, we calculate that the 57 instances of offer redemption were given \$1,178 in discounts. Therefore, from the redemption of the offers, the firm was able to obtain sales worth \$20,690 by spending \$1,178. By this measure, the profitability of the campaigns is not obvious, especially because the

Table 5. Average Observable Characteristics for Individuals in the Treatment and the Control Groups

	Control group		Treatment group		p -value
	Mean	Standard err.	Mean	Standard err.	
Total expenditure before the experiment	961.10	28.08	949.19	44.34	0.82
Number of offers redeemed before the experiment	0.0011	0.0002	0.0010	0.0007	0.94
Number of prior transactions	2.54	0.03	2.51	0.07	0.72
Number of prior experimental offers targeted	0.90	0.01	0.88	0.02	0.26
Number of prior treatments in the experiments	0.80	0.006	0.77	0.017	0.13
Number of purchases of baseball tickets	0.71	0.01	0.68	0.04	0.44
Number of purchases of other sports tickets	1.30	0.02	1.32	0.04	0.75
Population in the individual's area	27,413.13	148.33	27,172.64	361.52	0.54
% Black population in the individual's area	0.080	0.0011	0.078	0.0025	0.43
Avg. family size in the individual's area	3.164	0.0025	3.150	0.0061	0.03
Avg. household income in the individual's area	66,808.96	205.31	66,544.45	534.16	0.64

Notes. The last column shows the p -value of the test of equality of each characteristic across the groups. A joint test across all the factors yields a p -value = 0.33.

Table 6. Information on Transactions Involving Redeemed Offers

Number of individuals who get an offer	61,377
Number of transactions	1,475
Number of transactions redeeming an offer	57
Redemption rate:	
Number of transactions redeeming an offer	0.1%
Number of offers sent	
Expenditure in transactions with offer redemption	\$20,690
Estimate of the discounts given for the 57 redeemed offers	\$1,178

revenue attributed to redemptions may not be incremental: some individuals who redeemed the offers might have transacted on the platform even in the absence of the offers.

We were concerned that some users fail to apply the discount online and call later to avail the discounts. This could lead to an underestimation of the redemption rate observed in our data. To reassure ourselves, we contacted our data provider for information on postpurchase redemption. According to the company, only 0.253% of all discounts were later reimbursed over the phone during this time period (0.253% of 57 in expectation); i.e., these are instances where the buyer might have inadvertently missed including the discount code from a targeted promotion. Given its very small proportion, we do not think that our results are meaningfully affected by such instances.

To estimate the incremental sales caused by the offers, we next focus on the difference in sales between the experimental treatment and control groups.

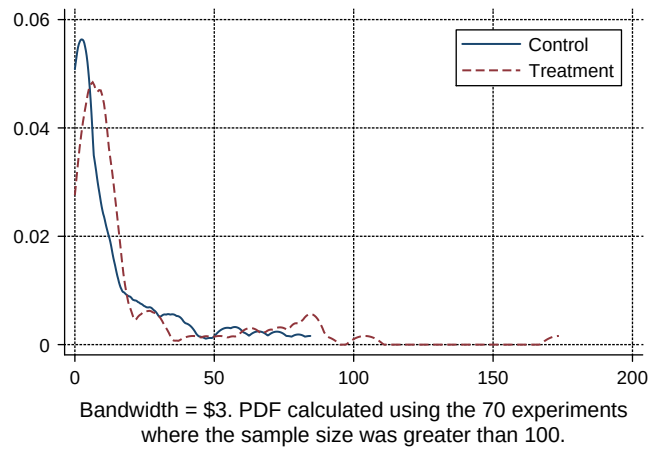
3.6. Raw Data Comparison of Treatment and Control Group Expenditures

We first check in the raw data, for any evidence of the impact of the experimental offers on consumers' spending across the experiments. We plot the distribution of average expenditures in the treatment and control groups across the 70 experiments in Figure 3. The plot shows that the distribution of the mean expenditures shifts to the right because of the experimental offers. This provides us with some reassurance that the effects of the offers are meaningful in a generalizable way across the experiments. In the rest of the paper, we present our approach and analysis to quantify the average causal effects across the offers and statistically test a variety of hypothesis relating to how they affect the consumer purchase behaviors.

4. Empirical Approach

As the platform's profits depend directly on users' expenditures, we aim to understand the impact of the email promotions on users' total expenditures during the period of validity of the promotion (denoted by Y) and to compare the difference in expenditures across

Figure 3. (Color online) Kernel-Smoothed Probability Density Function (PDF) of Average Expenditure in Treatment and Control Groups



treatment and control groups with costs associated with the discounts given. Before we proceed to the empirical analysis, we highlight the following points related to our empirical strategy.

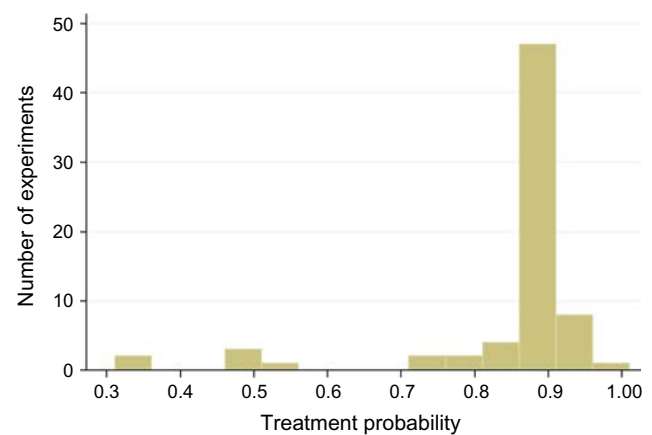
First, note that we cannot estimate the impact of these promotions on individuals who were never in an experiment run by the platform. Indexing individual users by i and experimental campaigns by j , we aim to estimate

$$\theta \equiv E_{ij}(Y_{ij}^1 - Y_{ij}^0 \mid D_{ij} = 1) \quad (1)$$

where Y_{ij}^1 and Y_{ij}^0 are the expenditures for i in the presence and the absence of an offer and D_{ij} is a dummy indicator of whether the individual i is a part of experiment j . In terms of the notation in the treatment effects literature, θ would be referred to as the average treatment effect on the treated population.

Second, the proportion of individuals who get allocated to the treatment group varies across experiments. Figure 4 shows the distribution of the treatment propensities across experiments. Note that the

Figure 4. (Color online) Distribution of Treatment Probabilities Across Experiments



treatment propensity across the 70 experiments varies significantly, ranging from 0.31 to 0.97 with an average of 0.85. This could be a consequence of the firm managing costs associated with experimentation. For example, if the target population is very receptive to offers, keeping a larger control group is more expensive because of the opportunity cost—foregone sales caused by sending no offers. Because of this variation in the data, we estimate θ by estimating a separate treatment effect (θ_j) for every experiment j , and compute a sample-weighted mean effect. Since every experiment j is a randomized experiment, comparison of the treatment and the control groups will give an unbiased estimate of the experiment-specific treatment effect on the population that was targeted for the experimental offer j . Mathematically,

$$\theta_j = E_i(Y_{ij}^1 - Y_{ij}^0 \mid D_{ij} = 1).$$

Suppose the sampling distribution of θ_j is estimated to be

$$\theta_j \sim N(\hat{\theta}_j, \hat{\sigma}_j^2).$$

Given the estimated sampling distribution of θ_j , we can obtain the sampling distribution of θ :

$$\theta = \frac{\sum n_j \theta_j}{\sum n_j} \sim N\left(\frac{\sum n_j \hat{\theta}_j}{\sum n_j}, \frac{\sum n_j^2 \hat{\sigma}_j^2}{(\sum n_j)^2}\right), \quad (2)$$

where n_j is the number of individuals in experiment j .¹³

4.1. Discussion

An alternative approach to estimating the average effect would be to pool data from all experiments and estimate a regression model with one coefficient for the treatment effect and fixed effects for all the experiments (e.g., Goldfarb and Tucker 2011). When the treatment proportion varies across experiments and is potentially correlated with the treatment effect, such an approach can lead to biased estimates. The appendix shows this possibility analytically.¹⁴ In a recent paper, Gibbons et al. (2014) discuss the importance of accounting for heterogeneous effects (as we do in our estimation) and estimating sample-weighted averages. They apply this approach to experimental papers published in the *American Economic Review* to highlight the difference it makes.

Second, we note that individuals in our data could be included in multiple experiments. This issue can potentially affect the standard errors of the estimated effects. However, this issue is important if the individuals in the control group in one experiment remain in the control group in other experiments, unlike in our case where individuals are rerandomized in every experiment. Still, we estimate the standard errors that are robust and clustered by individual, allowing for correlations in an individual's spending over time.¹⁵

5. Results

In this section we discuss our findings from the data. First, we focus on the impact of experimental offers on expenditure for targeted individuals during the time period when the offers were valid. If the offers are able to incentivize the individuals to purchase from the platform, we should see an increase in the expenditure for the individuals who get the offers relative to individuals who are excluded.

5.1. Expenditure While Offers Are Active

Following the approach highlighted in the previous section, we first compute the sampling distribution of the difference in the expenditure by individuals in the treatment group and the control group for every experimental campaign j . This effect is estimated by the ordinary least squares (OLS) estimate of the coefficient θ_j in

$$y_{ij} = \alpha_j + \theta_j T_{ij} + \epsilon_{ij}; \quad (3)$$

θ_j for all j are estimated jointly and the standard errors are robust, clustered by individual; T_{ij} indicates whether individual i was in the treatment group for experimental campaign j . We find that the $\hat{\theta}_j$ are jointly statistically significant across j ($p < 0.01$). We take an average of the sampling distribution of θ_j using (2) to get an estimate of θ , the effect of the experimental offers across campaigns on average. Table 7 shows the average effect. On average, the experimental offers cause the individuals who are targeted to spend \$3.03 more during the time when the offer is active. This estimate is statistically significantly different from zero.¹⁶ In other words, there exists evidence in the data suggesting that, on average, offers do work by increasing the expenditure and, consequently, revenues to the platform from buyers who received the offers relative to those from buyers who did not. Individuals in the control group spent \$8.07 on average. Therefore, the experimental offers caused a 37% increase in the spending on the website.^{17, 18}

Recall that the transactions in which offers are redeemed account for an expenditure of \$20,690. On

Table 7. Sampling Distributions of the Average Effect of the Experimental Offers on Dollar Expenditure (Difference Between Treatment and Control Groups) and of Control Group Expenditure for Relative Comparison

	Estimate	Std. err.
Avg. effect on expenditure across experiment offers	1.04 3.03**	
Avg. effect across experiment offers (ignoring transactions where an offer was redeemed)	2.81**	1.04
Avg. expenditure in the control group	8.07**	0.87

Note. $N = 71,445$.

** $p < 0.01$.

the other hand, the total increase in expenditures caused by the offers is significantly larger. It is $\$3.03 \times$ the number of offers sent = $\$185,972$, which is approximately 9 times the expenditure associated with offer redemptions. We repeat the calculation of the average effect of the experimental campaigns, ignoring the transactions where offers were redeemed, to compute $\theta_{\text{No redemption}}$. The estimate reported in Table 6 indicates that on average there is an increase of $\$2.81$ in the expenditure of individuals in the treatment group even when they do not redeem the offers. This analysis suggests that offers increase the expenditure for the recipients beyond the amount spent on the transactions where the offers are redeemed. Redemption of the offers does not provide a complete picture of the profitability of the offers. Therefore, from the perspective of the firm, the experimental campaigns are short-run profitable if an increase in buyer expenditure of $\$185,972$ justifies a spend of $\$1,178$ in giving discounts.

5.1.1. Description of the Effects. In this section we describe how the effectiveness varies across the 70 experimental campaigns. We first correlate the estimated effects of the offers with the characteristics of their target populations and other aspects of the offers, such as their respective durations and whether they are applicable only above a spending threshold. From the earlier discussion in Section 3.3 we know that the recency (r), frequency (f), and monetary value (m) of the past transactions made by an individual play an important role in targeting of offers. Therefore, we characterize the offers based on these characteristics of their target populations. For every experimental campaign we compute the average r , f , m of the target population and regress the estimated causal effect of the experimental offer on these factors. Column I of Table 8 shows the results from this regression. The coefficient for m is statistically significant and positive. This suggests that the offers targeted to individuals with a history of larger purchases with the websites lead to a larger increase in spending. Column II of Table 8 includes other aspects of the offers as explanatory variables in this regression. It shows that the other characteristics of the offers do not play a significant role in explaining the estimated effects. To describe the redemption of the offers in a similar manner, we regress the redemption rates of the offers on these target population characteristics. Columns III and IV of Table 8 show the estimates. We find that the average recency of past transactions plays a significant role. Specifically, the redemption rates of the offers decrease as the time since the last transaction increases for the targeted population. The other factors are not significantly associated with the redemption rates. Since the aspects of the campaigns such as characteristics of the target population (unobserved by us) are decided by the firm nonrandomly, we do not interpret these correlations as causal effects.

Table 8. Examining the Correlation Between the Estimated Treatment Effects Across 70 Experimental Offers and the Characteristics of the Offers and Their Targeted Population

	DV: Estimated causal effect of the experimental offer		DV: Redemption rate of the experimental offer (%)	
	I	II	III	IV
Target population's avg.: days since last transaction ($-r$)	−0.01 (0.01)	−0.01 (0.01)	−0.001** (0.0004)	−0.001** (0.0004)
Target population's avg.: number of past transactions (f)	−2.45 (4.73)	−2.80 (5.62)	−0.33 (0.23)	−0.42 (0.26)
Target population's avg.: value of past transactions (m)	0.02** (0.007)	0.02** (0.008)	0.0004 (0.0003)	0.0004 (0.0004)
Duration of the offer		0.07 (0.17)		0.01 (0.05)
Offer gives free delivery of tickets		8.55 (5.0)		−0.16 (0.20)
Offer is applicable above a spending threshold (0/1)		4.8 (7.6)		−0.21 (0.19)
Spending threshold level		−0.02 (0.04)		−0.001 (0.001)
Offer is applicable on a specific event		26.25 (23.35)		0.40 (0.34)

Note. $N = 70$.

** $p < 0.01$.

We now use the individual-level data to examine who among the individuals targeted with the offer is more likely to respond. Table 9 shows the estimates of (3) with the treatment indicator interacted with the RFM measures of the individuals. To allow for easy interpretation, we use the same quartile definitions for the r (days since the last transaction) and m (the monetary value of total prior transactions) measures as we did while describing the firm's targeting

Table 9. Examining the Correlation Between the Estimated Treatment Effects Across 70 Experimental Offers and the Characteristics of the Individuals, Conditional on Them Being Targeted by the Company

	DV: Expenditure while the offer is active	
Treatment interacted with		
2 ≤ Frequency of past transactions ≤ 3	2.66	(2.02)
4 ≤ Frequency of past transactions ≤ 9	10.43**	(3.68)
10 ≤ Frequency of past transactions	17.94	(22.37)
Total value past transactions: 2nd quartile	0.41	(1.39)
Total value past transactions: 3rd quartile	0.23	(1.57)
Total value past transactions: 4th quartile	3.80**	(1.88)
Days since last transaction: 2nd quartile	−0.38	(2.15)
Days since last transaction: 3rd quartile	12.29**	(1.35)
Days since last transaction: 4th quartile	11.73**	(1.26)
Intercept	−13.08**	(2.24)

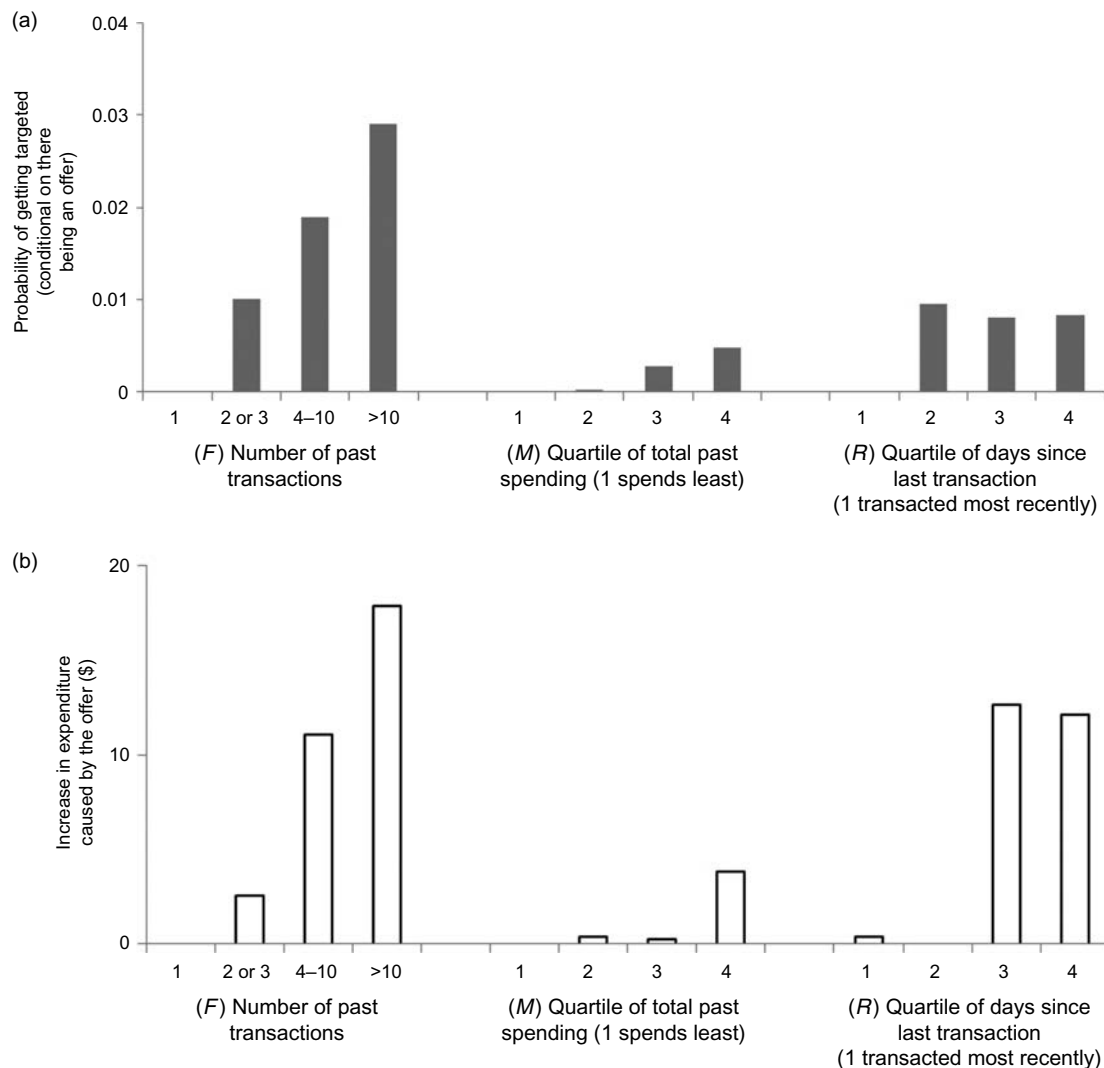
Note. $N = 71,445$.

strategy (in Table 4). The estimates show that the experimental offers have a higher effect on the spending of individuals who have made more transactions with the platform in the past. When we control for the number of past transactions, individuals who spend more also increase their spending more because of the offers. Interestingly, the offers are likely to have a higher impact on the individuals who did not transact recently on the website. It is consistent with the “reminder effect” of the experimental offers: offers remind individuals of the platform (or the product category), which drives them to make purchases on it.

Figure 5 shows the above heterogeneity in the estimated effectiveness of the offers together with the

probability of a segment getting targeted with an offer. Note that the targeting probability and the effectiveness of the offers both increase with f and m : the individuals who have spent more at the website are more responsive to the offers, and the company is more likely to target them with offers. However, this consistency between the targeting strategy and effectiveness of the offers is absent for r . The offers are least effective for individuals in the first and the second quartiles (those who transacted at the website relatively recently). However, the firm is most likely to send the offers to individuals in the second quartile. Although directionally the company seems to be following the strategy of not sending offers to the recent

Figure 5. Covariation of (a) the Probability of Getting Targeted and (b) the Average Effect of an Experimental Offer with Individual's Recency (R), Frequency (F), and Monetary Value (M) of the Past Transactions



Notes. (a) The bars in the top panel show the partial effect of changing one factor (of R , F , or M) on the probability of an individual getting targeted with an experimental offer, conditional on the company giving out offers. (b) The bars in the bottom panel show the partial effect of a factor holding others constant on the individual's expenditure (while the experimental offer is valid), conditional on being targeted with the offer. For example, individuals who transacted more than 10 times have a 3% chance of getting targeted with an offer. Conditional on being targeted, an offer causes an individual in this group to spend \$17 more than it causes an individual who transacted just once before when the offers were given out.

transactors, this analysis suggests that it could benefit more by sending more offers to buyers who are above the median in terms of days since they last transacted.¹⁹

So far, evidence presented in this section suggests that the experimental offer campaigns have an effect that is different from the traditional view on price reductions (Villas-Boas 1999, 2004). Price reductions are likely to increase purchases by low-valuation consumers by providing them direct incentives due to lower effective price. The consumers in our setting, however, are spending more because of the offers, without actually using the price discounts. The prominent mechanism driving the increase in expenditure in our context is consistent with a reminder effect and echoes the “advertising” effect of targeted email offers. We proceed by further exploring how the email offers impact the increase in expenditure.

5.2. Longer-Term Effects

The above analysis shows that the experimental offers, during the time that they are valid, cause an increase in expenditure on the website. The short-term increases in expenditure due to the experimental offers, however, may be offset by long-term decreases if consumers substitute across time by accelerating their purchases to be made during the offer window. Such purchase acceleration has been noted in other contexts in the marketing literature (e.g., Blattberg et al. 1995, Simester et al. 2009).²⁰ On the other hand, if the advertising-like effect is the driving mechanism, the offers may cause increases in expenditure in the longer term as well (Lewis and Reiley 2014, Sahni 2015).

If an offer serves as a reminder of the products available on the platform, its impact can decline depending

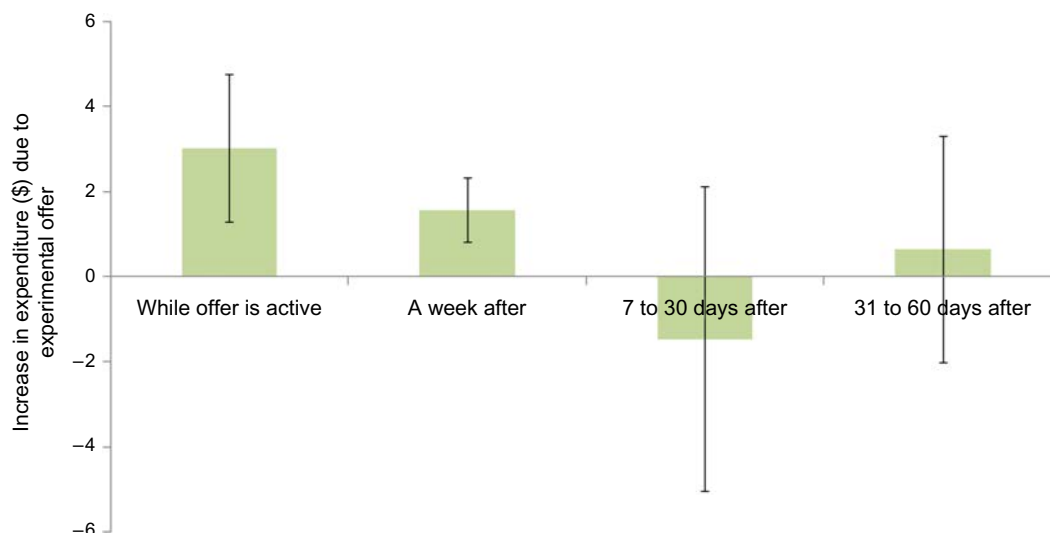
on the time elapsed from when it was sent (beginning of the offer) because of possible forgetting (Sahni 2015). However, its effect on expenditure at a point of time should not directly depend on whether the offer is valid. To check for this, we first examine if the offers cause individuals to spend more even after they expire. For this purpose, we compare the expenditures for individuals in the treatment and the control condition after the offers expire. The average increase in expenditures due to email offers is estimated using the same approach as before. For each experimental offer j we estimate the parameters of

$$\tilde{y}_{ij} = \alpha_j + \tilde{\theta}_j T_{ij} + \epsilon_{ij} \quad (4)$$

using OLS with robust standard errors clustered by individual, where $\tilde{y}_{ij}$ is i 's total expenditure during a period after offer j expires. We find that $\hat{\theta}_j$ are jointly statistically significant across j , indicating the presence of a long-term effect. Next, we compute the average of the sampling distribution of $\hat{\theta}_j$ to get the estimate of the average effect, again using (2).

Figure 6 shows the estimates and 90% confidence intervals for average changes in expenditure over periods of time while comparing between treatment and control groups across all the experiments. In the week after the offers expire, there is on average a \$1.55 increase in expenditure due to treatment. This estimate is statistically significantly greater than zero (p -value < 0.001). Therefore, even after the offer expires and can no longer be used to get discounts, it causes an increase in expenditure. This evidence is also consistent with the advertising effects of the email offers. Beyond a week after expiration, we do not find a

Figure 6. (Color online) Average Difference in Expenditure Between the Treatment and Control Groups Across Experiments During and After Offer Validity



Note. Error bars show 90% confidence intervals.

significant difference between the treatment and the control groups. So we do not find evidence that offers cause purchase acceleration.

The effects estimated above may be indirect effects of the offers. For example, offers may cause individuals to purchase while the offers are valid and then these individuals may repurchase in the future due to inertia or a positive consumption experience. In the data, we do not observe whether individuals in the treatment group actually remember the offers or spend more due to experiences while the offer is active. We can, however, observe how the carryover effects vary between individuals who transacted and individuals who did not during the offers. Therefore, we next estimate the interaction of the treatment with a dummy indicator, X_{ij} , which takes the value 1 if a transaction occurred during the period of the offer. This is implemented by estimating the coefficients for the following specification:

$$\tilde{y}_{ij} = \alpha_j + \delta_j T_{ij} + \beta_j T_{ij} \times X_{ij} + \gamma_j X_{ij} + \epsilon_{ij}. \quad (5)$$

We find that the coefficients are jointly significant across j . The estimated average coefficients are displayed in column III of Table 10. The coefficient on X_{ij} , which is an indicator of purchase while the offer was active, is positive and statistically significant. This is expected since individuals who transacted on the website in the past are likely to have a higher preference for the products offered and are therefore likely to spend more in the future. The interaction term is positive and significant too. This suggests that individuals who transacted while the offer was valid are more likely to spend in the week after the offer than individuals who transacted but did not receive the offer. Finally, the coefficient for the treatment term is also positive and marginally significantly different from zero. Individuals who received the offer but did not transact are spending approximately \$1.05 more than those who did not receive the offer due to the experiment. Therefore, there exists evidence suggesting that the direct effect of the experimental offers may persist and may

affect the consumers' propensity for purchasing at the platform over a period of time after offer expiration.

Next, we examine how the effectiveness of the offers decays over time from when the email is sent. For this purpose, we estimate the difference between the spending by the treatment and the control groups for every week for three months after the experimental offer is sent. Figure 7 shows this difference averaged across the experiments, along with the confidence intervals around it. It shows that the treatment group spends more on average than the control group for the first four weeks after the offer is sent. After four weeks, the difference is statistically insignificant.

5.3. Spillovers Across Categories

If the mechanism driving the effect of discount offers is through consumers being more likely to recall the website as a consumption option in the future, similar to an advertising effect, then the offers can affect sales of a broader set of products on the platform, and not just the products being promoted in the offers.²¹ In this section, we investigate whether genre-specific discounts impact the sales of tickets for other genres. Most of the observations in the data are not for offers that are specific to a genre. However, there exist many experimental offers that are not applicable for Major League Baseball (MLB) tickets. If the impact of the experimental offers is specific to the genre on discount, we could see substitution by individuals away from the excluded genre to the genre on sale. This would imply that, for offers that exclude MLB, we would expect to see a negative impact of these offers on expenditures on MLB tickets. On the other hand, if the offers broadly drive individuals toward the platform, we may find that these offers increase expenditure on MLB tickets, the excluded genre.

We have 51 offers and 25,154 observations of targeted buyers in campaigns that were not applicable for MLB. For this subset of offers, we estimate the effect of offer treatment on expenditure on MLB or on non-MLB tickets. Table 11 shows the estimates. Note that the

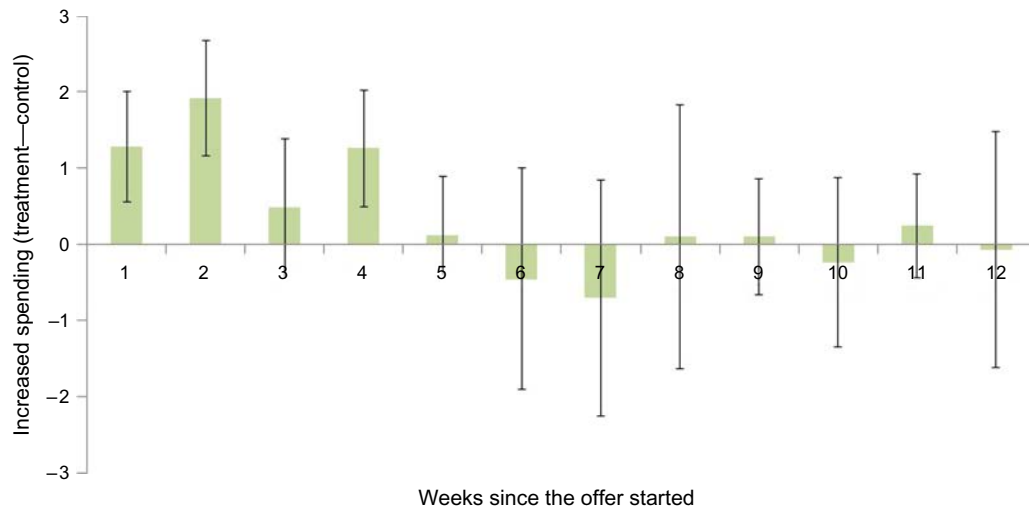
Table 10. Measuring the Impact of Experimental Offers on Expenditure in the Week After Experimentation

	DV: Expenditure in the week after the offer expires					
	I		II		III	
	Coeff.	Std. err.	Coeff.	Std. err.	Coeff.	Std. err.
Treatment	1.55**	0.45	1.65**	0.25	1.04**	0.22
Purchase during offer			24.37**	5.58	17.40*	9.08
Treatment × Purchase during offer					10.83*	5.62

Notes. The estimates are further split between buyers who purchased during the offer and those who did not. $N = 71,445$.

* $p < 0.05$; ** $p < 0.01$.

Figure 7. (Color online) The Effectiveness of the Offer (Spending in the Treatment Group—Control Group) Across Weeks from When the Offer Is Sent Out



Note. Error bars show 90% confidence intervals.

estimates are not as precise as before because the number of observations is one-third of the total observations in the data. In these data, getting the offer increases the expenditure by \$4.04 on average, as shown by column I in Table 11. Splitting these effects across non-MLB and MLB ticket purchases (columns II and III), we find that there is an increase in expenditure in both categories. We find an average increase of \$1.27 in expenditure on MLB tickets despite the fact that the offers were not applicable. This evidence indicates the presence of cross category spillover effects, similar to those documented in advertising (e.g., Erdem and Sun 2002).

These spillovers could exist because the offers draw individuals to the retail platform, but these individuals ultimately buy tickets for the genre they prefer. In this case, we expect the spillovers to MLB tickets to be driven by individuals who have a prior high propensity to purchase MLB tickets. To check for consistent evidence in the data, we further examine the characteristics of consumers who drive the estimated spillover effects. Specifically, we estimate a model where we interact offer treatment with an indicator of whether

the consumer purchased baseball tickets on the website in the past ($Past_MLB_{ij}$):

$$y_{ij} = \alpha_j + \delta_j T_{ij} + \beta_j T_{ij} \times Past_MLB_{ij} + \gamma_j Past_MLB_{ij} + \epsilon_{ij}. \quad (6)$$

We find that both sets of coefficients β_j s and δ_j s are jointly significant. Column IV of Table 11 shows that, on average, the interaction term's coefficient is positive and statistically significant. The main effect is still positive and marginally statistically significant. These estimates suggest that the experimental offers increase purchases of MLB tickets, mainly for individuals who have made a purchase in this genre in the past; these individuals increase their spending from \$1 to \$3 in this genre.

6. Discussion and Conclusion

This research improves our understanding of the effect of targeted emailed offers. Such marketing tools are typically viewed as levers for price discrimination. According to this view, a firm might want to send them

Table 11. Investigating the Treatment Effect for the Subset of Experimental Offers Which Were *Not* Applicable for MLB Tickets

	I		II		III		IV	
	Total expenditure		On non-MLB tickets		On MLB tickets		On MLB tickets	
	Coef.	Std. err.	Coef.	Std. err.	Coef.	Std. err.	Coef.	Std. err.
Treatment	4.04**	(1.96)	2.77	(1.89)	1.27**	(0.37)	0.33*	(0.17)
Treatment $\times$ (MLB purchase in the past)							3.10**	(0.36)
MLB purchase in the past							0.32*	(0.18)
Intercept	10.92**	(1.77)	9.81**	(1.68)	1.11**	(0.39)	0.70**	(0.12)

Notes. Further interacted with expenditures on MLB or non-MLB tickets, showing that genre spillovers are present. $N = 25,154$.

to individuals who are price sensitive and might use the discount to make purchases. Consequently, it suggests that the success of a campaign be measured by the redemption rate of the offers. We provide evidence against this view. We show that, in our setting, the offers increase consumer spending by 37.2%. However, this increase caused by the discount offers is not driven by the consumers using the discounts. A large fraction (90%) of the effect is unrelated to the redemption of the offers. Therefore, the offer redemption rate does not convey the complete value of the offer campaigns for the firm. Further, we examine the mechanism driving the effect and draw parallels with the advertising effect, which is not obvious a priori. We find that the effect of the discount offers carry over to after the offers expire. The effect also spills over to the genres that are not discounted by the offer. The effect is higher on individuals who have not transacted with the website recently. Therefore, the main effect of the offer emails is consistent with a reminder effect: the offers remind the consumers of the products available on the platform, which increases their spending on it. These findings suggest that the firms might want to target the offers to individuals who may have forgotten about their product offerings, rather than focusing specifically on price-sensitive individuals.

An implication of our research is that firms can gain from the targeted promotions even without having to incur the costs associated with the discounts. A natural question that arises is how these effects change with increased levels of promotion—the size of the discounts or repetition of the promotions. With our data it is hard to answer these questions because (1) we observe little variation in discount levels within an experiment, and (2) we also see little exogenous variation in the number of promotion emails received. The ability to vary promotion campaigns on these dimensions might allow future studies to examine further the advertising effect we find.

Our research also raises questions about how the impact of a price discount depends on the characteristics of the product or the consumers. Our data come from a ticket-selling website, which is different from a consumer packaged goods (CPG) product in a super-market setting (e.g., in terms of price or frequency of purchase), where most of the prior research has been conducted in the area of price discounts. Online buyers of tickets are also likely to have higher incomes. How such aspects of a market affect the significance of the advertising role of price discounts is an open question.

Another area of future research is the investigation of the impact of targeted discount offers on consumer search both within and across retail websites, thereby generating more detailed implications for firms. If offers reduce search across competing retailers, they might have implications for pricing for the platform.

At the same time offers might either inhibit or increase search on a retailer's own platform. While our findings on cross category spillovers shed some light on this aspect, detailed search data can potentially yield further insights.

Acknowledgments

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Appendix. Regression with Pooled Data

A possible alternate approach for analyzing data from multiple experiments is to pool all the observations and estimate one average treatment effect for all experiments, controlling for differences between experiments using fixed effects. In this section we show that this approach can lead to biased inference. Although each experiment is itself randomized, the experiment designs may be endogenous. For example, an experimenter may face different costs of treatment across different segments of consumers. As a result, the observed treatment proportion in an experiment may depend on the type of the target population, causing the treatment probability in an experiment to be correlated with the preferences of the target population.

To illustrate the bias, consider estimating the average treatment effect denoted by θ by estimating the following linear fixed effects model:

$$y_{ij} = \theta T_{ij} + \sum_{k=1}^J \gamma_k D_{ijk} + \epsilon_{ij}, \quad (7)$$

where a unit of observation is an individual i in an experiment j . T_{ij} is a dummy indicator of whether i was treated in j , and D_{ijk} is an indicator variable for $j = k$. The coefficient γ_k is the fixed effect for experiment k , and ϵ_{ij} represents the idiosyncratic factors. The OLS estimated coefficient is

$$\hat{\theta} = \frac{E(T'Qy)}{E(T'QT)}, \quad (8)$$

where $Q = I - D(D'D)^{-1}D'$ is the projection matrix to the space orthogonal to D . However, if the true model has experiment-specific sensitivities to T , then

$$\begin{aligned} y_{ij} &= \theta_j T_{ij} + \sum_{k=1}^J \gamma_k D_{ijk} + \epsilon_{ij} \\ \Rightarrow y_{ij} &= \bar{\theta} T_{ij} + \sum_{k=1}^J \gamma_k D_{ijk} + (\theta_j - \bar{\theta}) T_{ij} + \epsilon_{ij}. \end{aligned} \quad (9)$$

Combining (8) and (9) in the matrix form, we get

$$\begin{aligned}\hat{\theta} &= \frac{E(T'Q(T\bar{\theta} + D\gamma + T(\theta_j - \bar{\theta}) + \epsilon))}{E(T'QT)} \\ &= \bar{\theta} + \frac{E(T'QD)\gamma}{E(T'QT)} + \frac{E(T'Q(T(\theta_j - \bar{\theta})))}{E(T'QT)} + \frac{E(T'Q\epsilon)}{E(T'QT)} \\ &= \bar{\theta} + \frac{E(T'Q(T(\theta_j - \bar{\theta})))}{E(T'QT)}.\end{aligned}$$

The transition from the second to the third step follows because $QD = 0$ and $T'Q\epsilon = T'\epsilon = 0$ by definition of Q . Therefore the OLS estimate $\hat{\theta}$ from (7) provides a biased estimate of the average treatment effect $\bar{\theta}$, where the bias, $E(T'Q(T(\theta_j - \bar{\theta}))) / E(T'QT)$, is nonzero if the likelihood of treatment (given participation in an experiment) is correlated with the treatment sensitivities, θ_j , of the segments of individuals in each experiment.

The above discussion shows that the consistency of the estimates from the regression model (7) is dependent on whether the decision for the “degree” of experimentation is correlated with heterogeneity in treatment sensitivity across different experiments. Since in the data we observe some aggregate population characteristics for the areas where the individuals live, we can investigate the relation between these observed characteristics of the target population and the treatment proportions. For every experimental offer j we compute the average target population characteristics $\bar{X}_j$ and the proportion of the target population treated t_j . To understand the relationship between t_j and observed characteristics, we regress t_j on $\bar{X}_j$. Table A.1 shows the coefficients of the regression. Note that several estimated partial correlations are statistically significant. When the average household income of the target population increases, the size of the treatment group relative to the control group decreases. In such a case, where the experiment treatment proportion is correlated with characteristics of the population, it may also be correlated with treatment sensitivities of the individuals. Therefore, the above approach of pooling the data across experiments and using only experiment-level fixed effects as controls may lead to a biased estimated average treatment effect.

Table A.1. Relationship Between the Proportion of the Population Treated in an Experiment with the Observed Characteristics of the Population

	DV: Proportion of treatment in the experiment
% Black population	−9.99* (5.45)
Log avg. income	−0.42** (0.17)
Log population	−0.15 (0.13)
Avg. family size	−1.24** (0.39)
Intercept	10.8** (2.74)

Note. $N = 70$.

* $p < 0.05$; ** $p < 0.01$.

Endnotes

¹ Such possibilities have been explored in past research. We discuss how this paper relates to past research in Section 2.

² Email offers resemble offline coupons in that using the offer requires active redemption from the consumer’s side. However, they are easier to customize and match with the user’s preferences, allowing for more sophisticated targeting and cost management. Further, email offers can cause consumers to search immediately, because the websites with product information are just a click away. The resulting differences between targeted offers and coupons are evident in our data. First, compared to 38% of offline purchases in the cereal industry using coupons (Nevo and Wolfram 2002), we find that approximately 3% of transactions in our data actually use any offer. Second, we find that 90% of the short-term gains from the email offers is not through redemption of the offers, which is significantly different from what is found for offline coupons.

³ Other researchers have studied the role of the duration of the validity of coupons (Inman and McAlister 1994, Krishna and Zhang 1999).

⁴ We show later in Section 3.2 that individuals in our data who opt out of receiving emails are different on key dimensions from individuals who opt in and therefore are not suitable to be a control group.

⁵ We thank the Wharton Customer Analytics Initiative (WCAI) for providing us the data.

⁶ These offers are distinct from the “newsletter” emails used by our partner company (and others in e-commerce) to convey information about new events. The offers also do not provide explicit information about prices.

⁷ To better understand the individuals who buy on our partner company’s website, we used the comScore panel data set. We found that the buyers in our context are likely to have higher incomes, on average, relative to a typical buyer on the Internet. They are also more likely to be white and live in the northeast United States.

⁸ The actual number of people targeted by the campaigns could be larger than what we observe, because our data comprises only a random subsample of the population.

⁹ These numbers ignore the first transactions from each buyer, since buyers are not eligible for offers during their first transaction. The average is \$279 with a standard deviation of \$431 if the first transactions are included.

¹⁰ To check for the stability of the targeting strategy across periods in our data, we repeat the above analysis separately for the first and the second half of the data. Both the analyses show similar patterns as noted above.

¹¹ Past research has documented large variation in the redemption rates of offers/coupons across settings. For example, Reichhart et al. (2013) study emailed offers and find the response rate (number of individuals buying/number of individuals receiving coupons) to be 0.12%, which is very close to our estimate. In a recent paper on mobile coupons, Danaher et al. (2015) find redemption rates varying from 0.57% to 16%, with an average of 1%, in a setting where individuals actively seek coupons. This feature of their setting is different from ours and is likely to inflate the redemption rates.

¹² In cases where the individual had multiple promotional options within an offer, we assume that the person chose the best option to get an upper bound on the discounts given by the firm.

¹³ An implicit assumption here is that θ_j s are not correlated. We can relax this assumption, and do so in the empirical section to find no change in our inference.

¹⁴ The alternative approach that estimates a regression model with one coefficient for the treatment effect and fixed effects for all the experiments yields a higher (by 25%) estimate; the estimated effect is \$4 and is statistically significant. Therefore, our estimate of \$3.03 is more conservative.

¹⁵This issue is discussed in recent research in biostatistics. Lin and Sullivan (2009) recognize that repetition of individuals across experiments can affect estimates of standard errors and lead to an increase in false-positive inferences. Bhattacharjee et al. (2012) and Han et al. (2016) extend the discussion and prescribe adjustments to estimating standard errors. In the empirical section we follow their approach to check for robustness of our estimated precision.

¹⁶Relaxing the assumption about θ_j s being uncorrelated yields an estimate of \$3.03 with a standard error of 1.23 implying p -value = 0.01.

¹⁷As suggested in Equation (2), we weigh each experiment with the number of observations it comprises, to estimate an average effect per email offer. Estimating a simple average (without weights) yields an estimate of \$7.87 with a standard error of 3.17 (p -value = 0.01), which is larger than the estimated weighted average.

¹⁸For robustness, we reran our analysis after just keeping the first instance when an individual enters an experiment. This reduces our sample size by 40% (from 71,445 to 40,389) so we do expect loss of precision in the estimates. Still, this analysis yields an estimate of \$2.31 with a standard error of 1.36, which is marginally statistically significant (p -value < 0.1). This estimate is statistically indistinguishable from our average estimate of \$3.03. Additionally, we follow the approach suggested by Han et al. (2016) and estimate the standard errors using their suggested methodology (outlined on p. 10 of that paper). As expected, we find little change in the estimates: average effect of the offer is estimated to be \$3.01 with a standard error of 0.91 (p -value < 0.01).

¹⁹This analysis just focuses on spending on the website while the offer is active. The company might see other benefits (word of mouth) that we do not measure. Additionally, the company might have other information for segmentation (apart from RFM variables) that are unobserved by us and could factor into their strategy.

²⁰Other interactions between the treatment and the control groups could also cause a short-term difference in behaviors of the treatment and control groups, especially if the dependent measure is narrowly defined (e.g., Blake and Coey 2014).

²¹For example, such effects have been documented by Sahni (2016) for online advertising. Anderson and Simester (2013) show spillovers to competitors in the context of catalog mails.

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